

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-I

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Control System Basic Structure, Block Diagram Reduction and Signal Flow Graphs

Name of the Innovative Practice: Zero Minute Speech, Collaborative Study and Minute Paper

Date & Duration: 01.07.2019 (10 Minutes), 08.07.2019 (25 Minutes) & 10.07.2019 (10 Minutes)

Description:

(i) Zero Minute Speech:

Zero Minute Speech will lead to express basics of the particular topics by different students. The important topics were allotted to set of students and they will give speech on the topic with doubt clarification.

(ii) Collaborative Study:

Collaborative learning is a situation in which two or more people learn or attempt to learn together. Unlike individual learning, people engaged in collaborative learning capitalize on one another's resources and skills.

The following are smart collaborative studying strategies for all students.

1. Understand that collaborative studying can be more effective.
2. Establish the core tenets of your group.
3. With core tenets in place, tackle the problem at hand.
4. Be ready to use online methods of communication.
5. Incorporate strategic contingency plans as problems arise.

(iii) Minute Paper:

- Minute papers can provide a “conceptual bridge” between successive class periods. For instance, at the beginning of class, a quick review of student responses to a minute paper answered at the end of a previous class can provide an effective segue between successive class sessions.
- Minute papers can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.
- Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally.

Some of most frequently used minute-paper category:

1. Interest.
2. Relevance.
3. Analysis.
4. Conceptual Connections.

Goals (Learning Outcomes):

1. The students will be able to analyse basic components of control system.
2. The students can be acquired better knowledge about block diagram techniques.
3. The students can be easily identified the transfer function of the system using signal flow graphs models.

Use of appropriate method:

Justification for choosing the Activity:

1. **Zero Minute Speech:** Leads to express basics of the particular topics by different students.
2. **Collaborative Study:** Develops higher level thinking skills among the students and Stimulates critical thinking and helps students clarify ideas through discussion and debate
3. **Minute Paper:** The great advantage of Minute Papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

1. **Zero Minute Speech:** I choose this tool because it does not need a big preparation so I can involve the slow-learners also.
2. **Collaborative Study:** Development of higher-level thinking, oral communication, self-management, and leadership skills.
3. **Minute Paper:** The great advantage of Minute Papers is that they provide manageable amounts of timely and useful feedback for a minimal investment of time and energy.

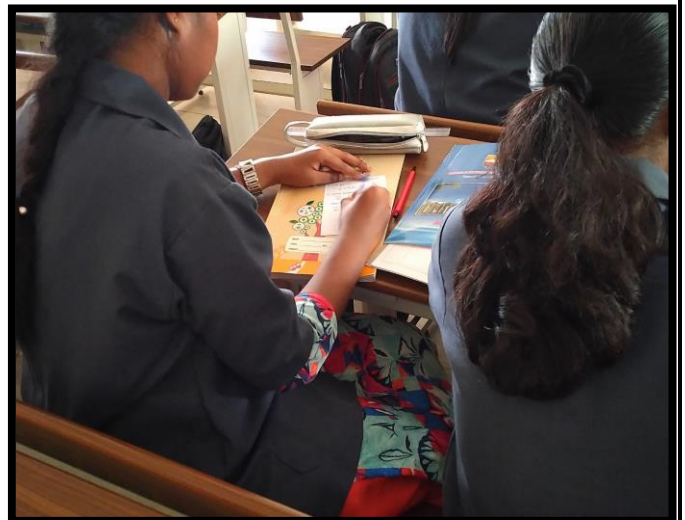
Innovative Teaching Method Execution

Zero Minute Speech for the Topic Control System Basic Structure on 01.07.2019



Collaborative Study for the Topic Block Diagram Models - Reduction Procedure on 08.07.2019





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ECE - A.

① $T = \frac{1}{\Delta} \sum_k P_k \Delta_k$

② Positive - $\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$

Negative - $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

③

$i(t) = \frac{v(t)}{R}$

$v(t) = L \frac{di(t)}{dt}$

$i(t) = C \frac{dv(t)}{dt}$

953618106006

2. Overall gain, $T = \frac{1}{\Delta} \sum P_k \Delta_k$

by For positive feedback $\frac{C(s)}{R(s)} = \frac{G(s)}{1 - G(s)H(s)}$

For negative feedback, $\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$

cf $\frac{R}{s}$ $v(t) = Ri(t)$ $i(t) = \frac{v(t)}{R}$

$\frac{1}{s}$ $v(t) = \frac{1}{s} \frac{di(t)}{dt}$ $i(t) = \frac{1}{s} \int v(t) dt$

$\frac{1}{s^2}$ $v(t) = \frac{1}{s^2} \int i(t) dt$ $i(t) = C \frac{dv(t)}{dt}$

Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ✎ **Zero Minute Speech:** Few students brief the control system basic structure and interacted with their class mates.
- ✎ **Collaborative Study:** Students are able to learn more material by engaging with one another and making sure everyone understands, students retain more information from thoughtful discussion, and students have a more positive attitude about learning
- ✎ **Minute Paper:** Helps the students to recall the examples for each addressing modes.

Changes required for future (if required):

- Need to give chance to all students for zero minute's speech.
- Extra hour need to plan for collaborative Study.

Reflective Critique:

Challenges:

1. **Zero Minute Speech:** Not able to give chance to all the students
2. **Collaborative Study:** Group cohesiveness occurs only to the extent that people's needs are satisfied.
3. **Minute Paper:** Not able to involve all the students.

Benefits:

1. **Zero Minute Speech:** Few students share their ideas and information with classmates.
2. **Collaborative Study:** Collaborative learning makes classroom management easier, when teachers assign tasks to students in groups, they can monitor progress by checking in on five or six different groups instead of all individual students.

3. **Mind Map:** Provide a “conceptual bridge” between successive class periods and involves all students in class simultaneously.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	1			1	1	1	1	1

CO1: The students will be able to perform modeling of control system using various techniques.

References:

1. <https://franticallyspeaking.com/1-minute-speech/>
2. <https://teaching.cornell.edu/teaching-resources/engaging-students/collaborative-learning>
3. <http://www.inspiration.com/visual-learning/mind-mapping>
4. A.Nagoor Kani, “Control Systems Engineering”, RBA Publications, First Edition, 2013.

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-II

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Transient Response & Steady State Response and PD, PI, PID control systems

Name of the Innovative Practice: Simulation Tool & Mind Map

Date & Duration: 29.07.2019 (20 Minutes) & 03.08.2019 (20 Minutes)

Description:

(i) Simulation Tool:

- ❖ Simulation software is based on the process of modelling a real phenomenon with a set of mathematical formulas. It is, essentially, a program that allows the user to observe an operation through simulation without actually performing that operation.
- ❖ A primary advantage of simulators is that they are able to provide users with practical feedback when designing real world systems. This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed.

(ii) Mind Map:

- ❖ A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.
- ❖ A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

- ❖ The students will be able to obtain the time response and steady state error of control systems using simulation tool.
- ❖ The students will be able to analysis the system stability using different compensator.

Use of appropriate method:

Justification for choosing the following Activities:

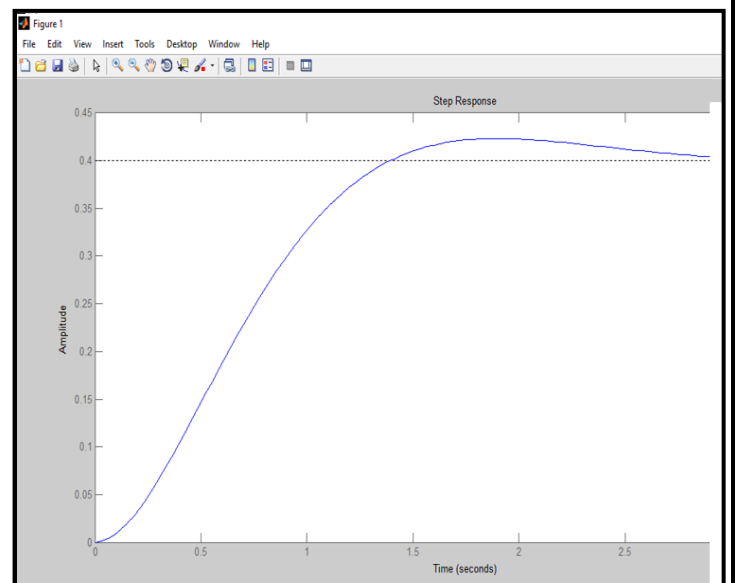
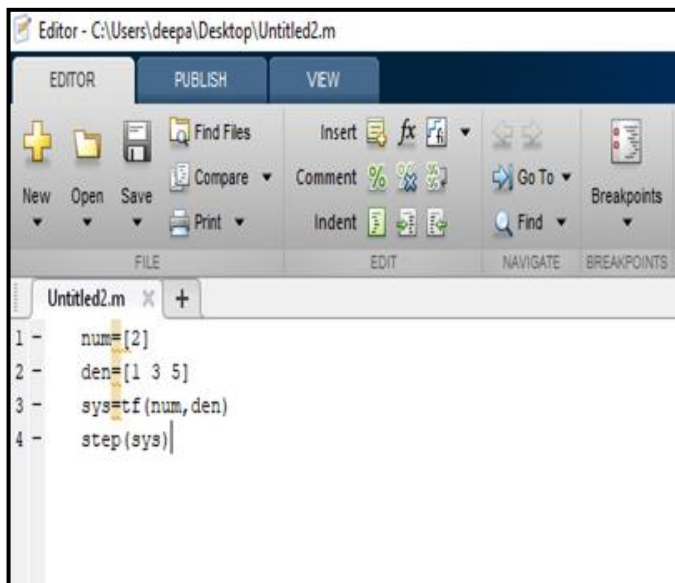
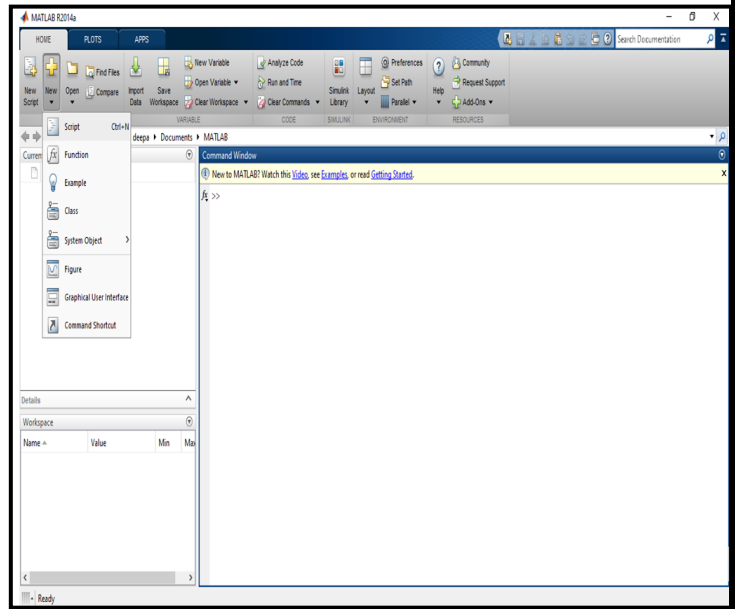
- ❖ **Simulation Tool:** It provides users with practical feedback about systems transient responses.
- ❖ **Mind Map:** Help students get a better understanding of new ideas by having them create a mind map. A mind map can assist with understanding because it conveys hierarchy and relationships, allowing students to see the big picture.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

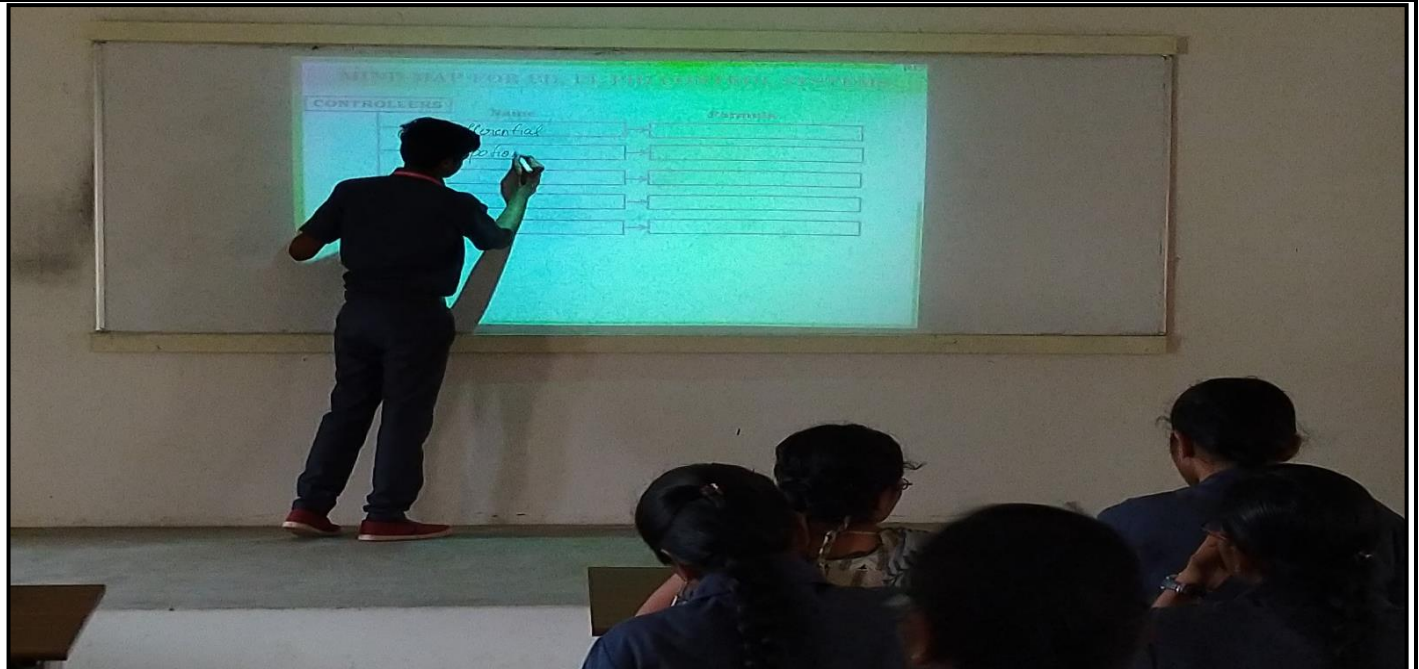
- ❖ **Simulation Tool:** This activity requires more time to complete because need to understand the step by step procedure to get the result, so not able to complete with in the scheduled time because time took to deliver the procedures.
- ❖ **Mind Map:** Usually Students took time to complete group related tasks, so I asked the students to prepare with few questions and I explained the how to do this activities with some examples.

Innovative Teaching Method Execution

Simulation Tool for the Topic Transient Response & Steady State Response on 22.07.2019



Mind Map for the PD, PI, PID control systems on 05.08.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ❖ **Simulation Tool:** Students understand the system response for the given input without manual calculation and he tried for other systems also.
- ❖ **Mind Map:** Because of group work, they shared their view though this activity among the friends with full interest.

Changes required for future (if required):

- ❖ To complete the activity within the time, the student requires some basic idea about the activities before one day so I need plan accordingly in future.

Reflective Critique:**Challenges:**

- ❖ **Simulation Tool:** Few students were not able to follow the procedures in the software.
- ❖ **Mind Map:** In few situations, students struggled to complete their task, so I gave some hints to complete the activity.

Benefits:**Simulation Tool:**

- ❖ One of the primary advantages of simulators is that they are able to provide users with practical feedback when designing real world systems.
- ❖ This allows the designer to determine the correctness and efficiency of a design before the system is actually constructed.

Mind Map:

- ❖ Help students brainstorm and explore any idea, concept, or problem.
- ❖ Facilitate better understanding of relationships and connections between ideas and concepts.
- ❖ Make it easy to communicate new ideas and thought processes.
- ❖ Allow students to easily recall information.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	3	1	1	1	-	-	1	1	1	1	1

CO2: *The students will be able to obtain the time response and steady state error of control systems.*

References:

- ❖ <https://www.sdera.wa.edu.au>
- ❖ <http://www.inspiration.com/visual-learning/mind-mapping>
- ❖ A.Nagoor Kani, "Control Systems Engineering", RBA Publications, First Edition, 2013.
- ❖ <https://ch.mathworks.com/help/control/ref/lsim.html>
- ❖ <https://ch.mathworks.com/solutions/control-systems.html>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-III

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Bode Plot and Polar Plot

Name of the Innovative Practice: Audio & Video Tools and Buzz Session

Date & Duration: 08.08.2019 (25 Minutes) & 19.08.2019 (25 Minutes)

Description:

(i) Audio & Video Tools:

- ✓ According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- ✓ Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- ✓ Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

(ii) Buzz Session:

- ✓ A buzz group is a small, intense discussion group usually involving to 3 persons responding to a specific question or in search of very precise information. The full plenary group is subdivided into the small groups. It's called a 'buzz' group because it mimics the sound of people in intense discussion.
- ✓ In such sessions, small sub-groups of two to four persons spend a short period (generally no more than five minutes) intensively discussing a topic or topics suggested by the teacher. Each sub-group then reports back on its deliberations to the group as a whole, or sometimes combines with another sub-group in order to share their findings and discuss the implications.

Goals (Learning Outcomes):

- ✓ The students will be able to acquired better knowledge about Bode Plot and Polar Plot.
- ✓ The students will be able to analysis the system phase margin and gain margin for the system.

Use of appropriate method:

Justification for choosing the following Activities:

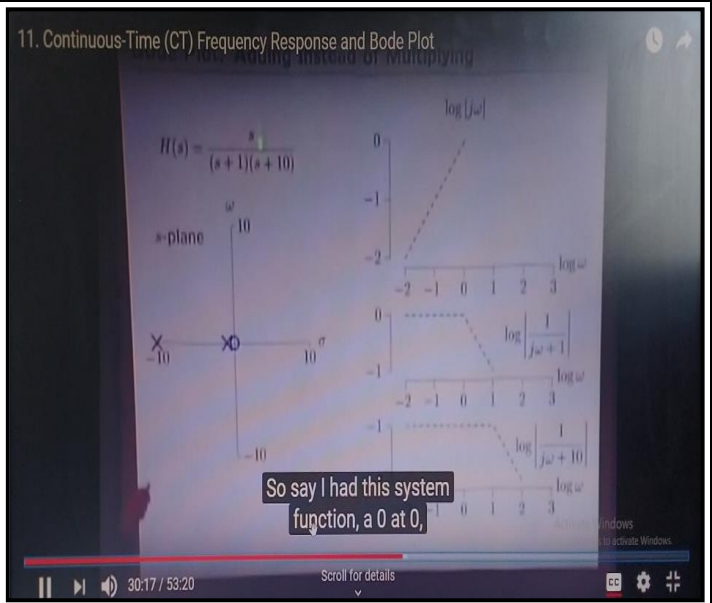
- ✓ **Audio & Video Tools:** According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- ✓ **Buzz Session:** Buzz sessions are short participative sessions that are deliberately built into a lecture or larger group exercise in order to stimulate discussion and provide student feedback.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

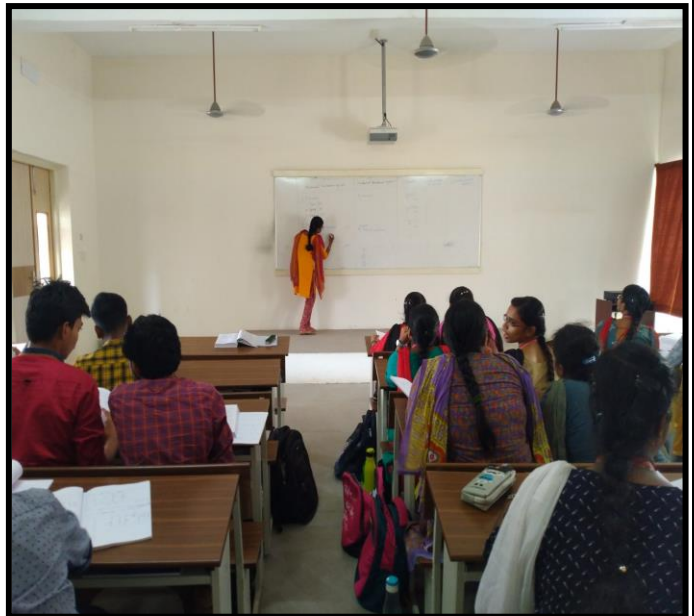
- ✓ **Audio & Video Tools:** Sometimes audio video tool were lengthy session, so I choose the video clips which more essential to the students.
- ✓ **Buzz Session:** First I explained the benefit of this activity and then I created the groups based on their choices, so they were very happy to mingle with the friends to do this activity.

Innovative Teaching Method Execution

Audio & Video Tools for the Topic Bode plot on 14.08.2019



Buzz Session for the Topic Polar Plot on 05.08.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ✓ **Audio & Video Tools:** With audio & video presentation the students got different types of procedures and techniques.
- ✓ **Buzz Session:** The small groups discussed the topics and they came with interesting solutions to remember and solve the problems.

Changes required for future (if required):

- ❖ Need to pick best Video section with shorter delivery duration with clear audio.
- ❖ Topics for activities are students choice in future will give better result.

Reflective Critique:**Challenges:**

- ✓ **Audio & Video Tools:** Sometimes audio video tool were lengthy session.
- ✓ **Buzz Session:** Lack of participation by all group members.

Benefits:

- ✓ **Audio & Video Tools:** Constitute an excellent method of introducing variety into a lecture, thus helping to maintain student attention. Audio video tool gives different teaching experience to the students with some animation.
- ✓ **Buzz Session:**
 - Highly participative.
 - Very focussed and direct.
 - Very frequently it integrates theory with experience.
 - Short, intense and using trainees own information so there is ownership of the output by trainees.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	3	1	1	1	-	-	1	1	1	1	1

CO3: The students will be able to design various compensators and to analyse the frequency response of the system using various plots.

References:

- ❖ <https://www.youtube.com/watch?v=2X7o37pfdp8>
- ❖ <http://www.kstoolkit.org/buzz-groups>
- ❖ A.Nagoor Kani, "Control Systems Engineering", RBA Publications, First Edition, 2013.
- ❖ <http://www2.rgu.ac.uk/celt/pgcerttlt/selecting/select18.htm>
- ❖ <http://sabarishedn.blogspot.com/2015/06/techniques-of-teaching-science-brain.html>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-IV

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Routh stability criterion & Root locus

Name of the Innovative Practice: Zero Minute Speech & Audio & Video Tools (MIT Open Courseware)

Date & Duration: 12.09.2019 (5 Minutes) & 19.09.2019 (15 Minutes)

Description:

(i) Zero Minute Speech:

- ✿ Zero Minute Speech will lead to express basics of the particular topics by different students.
- ✿ The important topics were allotted to set of students and they will give speech on the topic with doubt clarification.

(ii) Audio & Video Tools:

- ✿ According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.
- ✿ Use of audio-visual aids help in maintaining discipline in the class since all the students' attention are focused in learning. This interactive session also develops critical thinking and reasoning that are important components of the teaching-learning process.
- ✿ Students learn when they are motivated and curious about something. Traditional verbal instructions can be boring and painful for students. However, use of audio-visual provides intrinsic motivation to students by peaking their curiosity and stimulating their interests in the subjects.

Goals (Learning Outcomes):

- ✿ The students will be able to determine the stability of control systems using Routh stability criterion.
- ✿ The students will be able to obtain the system performance using Root Locus.

Use of appropriate method:

Justification for choosing the following Activities:

- ✿ **Zero Minute Speech:** Students can expand their thinking abilities. Zero minute impromptu speeches can inspire, provoke, and persuade the listeners.
- ✿ **Audio & Video Tools:** According to the Webster dictionary, audio-visual aids are defined as "training or educational materials directed at both the senses of hearing and the sense. Audio-visual provides opportunities for effective communication between teacher and students in learning.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- ✿ **Zero Minute Speech:** I choose this tool because it does not need a big preparation so I can involve the slow-learners also.
- ✿ **Audio & Video Tools:** Sometimes audio video tool were lengthy session, so I choose the video clips which more essential to the students.

Innovative Teaching Method Execution

Zero Minute Speech for the Topic Root stability criterion on 12.09.2019



Audio & Video Tools (MIT open courseware) for the Topic Root locus on 19.09.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ❁ **Zero Minute Speech:** Each student will deliver their thoughts related to the topic and clears doubts along with the friends.
- ❁ **Audio & Video Tools:** With audio & video presentation the students got different types of procedures and techniques.

Changes required for future (if required):

- ❖ Need to give topics to the students to choose among them before two or three days to complete the activity within the scheduled time.

Reflective Critique:

Challenges:

1. **Zero Minute Speech:** Not able to give chance to all students during zero minute speech practice.
2. **Audio & Video Tools:** Not able to make students more interactive.

Benefits:

1. **Zero Minute Speech:** Few students share their ideas and information with classmates and Some students interacted with other students to clarify the doubts.
2. **Audio & Video Tools:** Audio video tool gives different teaching experience to the students with some animation.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	1	1	-	-	1	1	1	1	1

CO4: The students will be able to determine the stability of control systems.

References:

- ❖ https://www.youtube.com/watch?v=BGug_49cGYw
- ❖ <https://www.presentationblogger.com/3-simple-steps-how-to-do-a-speech-on-any-topic-with-zero-preparation/>
- ❖ M.Gopal, “Control System – Principles and Design”, Tata McGraw Hill, 4th Edition, 2012.
- ❖ A.Nagoor Kani, “Control Systems Engineering”, RBA Publications, First Edition, 2013.
- ❖ <http://www.icaltefl.com/just-a-minute-speaking-activity>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)
Innovative Practices Description for Unit-V

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Mr.B.Kannan

Name of the Topic: Controllability and Observability & Digital control design using state feedback

Name of the Innovative Practice: Exit Slips & Brain storming

Date & Duration: 30.09.2019 (10 Minutes) & 05.10.2019 (25 Minutes)

Description:

(i) Exit Slips:

- ✦ They provide faculty member with an informal measure of how well students have understood a topic or lesson.
- ✦ They help students reflect on what they have learned.
- ✦ They allow students to express what or how they are thinking about new information.
- ✦ They teach students to think critically.
- ✦ At the end of your lesson ask students to respond to a question or prompt.
 - Prompts that document learning:
Example: Write one thing you learned today.
Example: Discuss how today lesson could be uses in the real world.
 - Prompts that emphasize the process of learning:
Example: I didn't understand...
Example: Write one question you have about today's lesson.
 - Prompts to evaluate the effectiveness of instruction:
Example: Did you enjoy working in small groups today?

(ii) Brain storming:

Brainstorming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members.

Withhold criticism: In brainstorming, criticism of ideas generated should be put 'on hold'. Instead, participants should focus on extending or adding to ideas, reserving criticism for a later 'critical stage' of the process. By suspending judgment, participants will feel free to generate unusual ideas.

Welcome wild ideas: To get a good long list of suggestions, wild ideas are encouraged. They can be generated by looking from new perspectives and suspending assumptions. These new ways of thinking might give you better solutions.

Goals (Learning Outcomes):

- ✦ The students will be able to express the information verbally about Controllability and Observability.
- ✦ The students will be able to analysis Digital control design using state feedback by group.

Use of appropriate method:

Justification for choosing the following Activities:

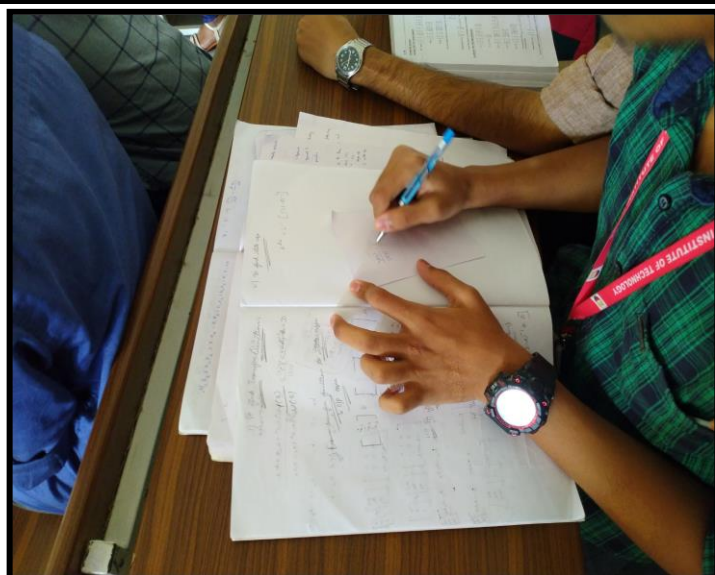
- ✦ **Exit Slips:** It allow the faculty member to measure understanding level of the students.
- ✦ **Brain Storming:** This creates more opportunities to enhancing divergent solution

Effective Presentation: (Implementation (Plan & Execution) with Proof):

- ✦ **Exit Slips:** It allow the faculty member to measure understanding level of the students. I gave brief introduction about how to write the exit slip with more informative.
- ✦ **Brain Storming:** It is a group activity so students were selected their group members themselves and started discuss about the topic finally they came with new ideas and information's.

Innovative Teaching Method Execution

Exit Slips for the Topic Controllability and Observability on 30.09.2019



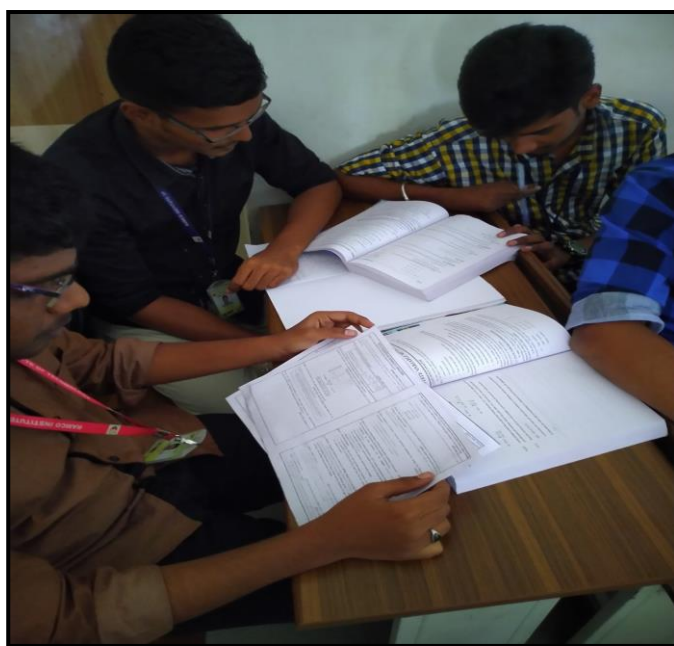
EXIT SLIP

$$\frac{Y(s)}{U(s)} = C(SI - A)^{-1}B + D$$

$$e^{At} = L^{-1}[(SI - A)^{-1}]$$

$$(SI - A)^{-1} = \frac{\text{cofactor of } SI - A}{\text{Determinant of } SI - A}$$

Brain storming for the Topic Digital control design using state feedback on 05.10.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ✦ **Exit Slips:** Within a time they can recall the specific topic and put it in the exit slip, which leads to remember the topic for long time.
- ✦ **Brain Storming:** Each group members tried their best and finally came with suitable message for the given topic and delivered to the friends.

Changes required for future (if required):

- ✦ Time management is very important to complete the activity effectively, so I need to prepare topics and groups based on the student's willingness before a day.

Reflective Critique:

Challenges:

- ✦ **Exit Slips:** Not all students actively participated.
- ✦ **Brain Storming:** Requires more time to complete because they took much time for discussion.

Benefits:

- ✦ **Exit Slips:**
 1. Prompts that document learning.
 2. Prompts that emphasize the process of learning.

Brain Storming:

1. Reduce social inhibitions among group members.
2. Stimulate idea generation.
3. Increase overall creativity of the group.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	2	1	1	1	-	-	1	1	1	1	1

CO5: The students will be able to analyse and obtain state space model using state variables.

References:

-  <https://www.youtube.com/watch?v=up78Mm74ag0>
-  John.B.Peatman, "Design with PIC Microcontroller," Prentice Hall, 1997.
-  https://www.readingrockets.org/strategies/exit_slips
-  <http://www.adlit.org/strategies/19805/>